Sahasra Reddy Peddakkagari

me24btech11051@iith.ac.in | sahasrareddy0906@gmail.com | +91 7396811187 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.28
NARAYANA JUNIOR COLLEGE	Class XII (Telangana state board of intermediate education)	2024	97.50%
MATRUSRI DAV PUBLIC SCHOOL	Class X (CBSE)	2022	93.60%

SKILLS

Programming: C, C++, Python, SQL

CAD & Engineering: Solid Edge, MATLAB, Abaqus, LTspice

Web Development: React, FastAPI, HTML, CSS

Databases: PostgreSQL, Supabase

WORK EXPERIENCE

Software Development Intern | CBAI Technologies Pvt. Ltd.

- Developed a **full-stack document processing platform** using React, Django and REST APIs, automating document upload, processing and result management.
- Optimized backend processing through **parallel execution** to reduce processing time for batch workloads.
- Containerized application services using **Docker** and deployed backend components on **AWS** for scalable processing.
- Designed and integrated **PostgreSQL** data models for document metadata, processing jobs and results.

Tech Stack: Python, React, Django, REST APIs, PostgreSQL, Docker, AWS, Git

PROJECTS

Compact Linkage Mechanism Design

- Designed a compact linkage mechanism in Solid Edge to achieve the required range of motion while minimizing overall assembly size.
- Modeled individual components and created a fully constrained assembly with proper mating relationships.
- Optimized the linkage geometry to eliminate interference and ensure smooth motion throughout the operating cycle.
- Performed tolerance analysis and adjusted retaining ring clearances to improve assembly reliability.
- Incorporated standard mechanical components such as shafts, bearings, retaining rings, and fasteners into the assembly.
- Used parametric CAD modeling to enable quick design modifications and iterative optimization.
- Designed print-in-place CAD components to simplify assembly and improve manufacturability.

AI Recruitment and Online Assessment Platform

- Developed a full-stack recruitment platform using React, FastAPI, and Supabase for end-to-end hiring workflows.
- Implemented candidate registration, online assessments, automated question assignment, and evaluation modules.
- Designed a PostgreSQL database schema with secure authentication and RESTful APIs.
- Designed an AI-assisted assessment workflow for automated candidate evaluation and recruitment support.
- Developed RESTful backend services for candidate management, assessment workflows, and database integration.

3D Printable JCB Model

- Designed a miniature JCB model optimized for additive manufacturing.
- Improved component assembly and printability through design modifications.
- Created a functional multi-part CAD assembly for 3D printing.

POSITIONS OF RESPONSIBILITY

Internship & Placement Cell Coordinator

- Coordinated with recruiters for internship and placement activities.
- Assisted in organizing pre-placement talks, recruitment drives, and corporate interactions.
- Managed communication between recruiters, students, and the placement office.
- Supported logistics and smooth execution of campus recruitment processes.

Coordinator

- Coordinated activities and events involving international students and delegates.
- Assisted in communication between international students and institute administration.
- Supported planning and execution of institute-level international programs.

Coordinator

- Coordinated practice sessions, performances, and team activities.
- Managed communication and scheduling among team members.
- Assisted in organizing institute cultural events and dance performances.

EXTRACURRICULAR

Represented **IIT Hyderabad** in the **Group Dance** event at **Inter IIT Cultural Meet 8.0**, collaborating as part of the institute's cultural contingent.

Member of the **National Service Scheme (NSS)**, actively participating in community service and social welfare initiatives.